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Analyzing Safety Risk Imposed by Jobsite Debris to Nearby Built Environments Using Geometric Digital Twins and Vision-Based Deep Learning

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Abstract: Extreme wind events can pick up loose and small objects on the ground, and once the objects become airborne, they negatively impact surrounding communities due to the collision impact. In this regard, jobsites and laydown yards that involve construction materials such as gravel piles and crushed rocks could be the main sources of potential windborne debris during extreme wind events. To analyze safety risk and predict the damage imposed by jobsite debris to nearby built environments, a new computer vision-based risk assessment based on geometric digital twins of jobsite debris is proposed for the reliability analysis on glazing systems of dwellings located on nearby jobsites. The impact of a gravel pile in a railroad jobsite on nearby buildings and residential environments was studied based on extreme wind event scenarios, and the failure risk of the building glazing system was computed. The risk associated with jobsite debris during extreme wind events and their impact on neighboring communities are analyzed through three computing modules: (1) satellite imagery-based terrain modeling to study 3D characteristics of the at-risk built environment; (2) analyzing visual data from Google Street View to assess the risk associated with glazing panels of dwellings in the communities; and (3) analyzing visual data from a jobsite to quantify the impact of jobsite debris, to associate its safety risk to neighboring communities. The proposed method can provide an immediate heads up for those who reside nearby jobsites, allowing to take required preemptive actions to protect their habitation against potential windborne debris. Practitioners will also be informed of such jobsite debris-related risk before extreme wind events to better secure their jobsites for the risk mitigation.
DOI: 10.1061/(ASCE)CP.1943-5487.0001044. *© 2022 American Society of Civil Engineers.*

Introduction

Natural disasters such as hurricanes impose a great risk to the built environment, causing billions of dollars worth in damage each year. The primary victims are the communities and neighborhoods that are woefully unprepared for the disasters. For example, hurricane damages can range from structural damages (Pinelli et al. 2004) to flooding and inundation due to heavy rain (Alizadeh and Behzadan 2021). Particularly, the majority of damage in extreme wind events are imposed due to the collision impact of potential windborne debris (PWDs) (Beason et al. 1984; Doddipatla and Kopp 2019; Vickery et al. 2006). As a poignant example, the aftermath of Hurricane Alicia imposed substantial window glass damage in the downtown Houston area (Beason et al. 1984). On-site investigation identified that windborne missiles (e.g., gravel) substantially contributed to the amount of glazing damage and window breakage, especially to high-rise towers (Beason et al. 1984). The gravels in Hurricane Alicia were sourced from roofs of nearby buildings, and caused damage to more than 5,700 glass panes that were eventually replaced (Beason et al. 1984). To minimize the risk of such airborne PWDs during extreme wind events, the Federal Emergency Management Agency (FEMA) of the United States Department of Homeland Security has leveraged a physical-based risk model to quantify the risk of glazing damage for buildings and residential environments, and the model has been used to carry out the reliability analysis for glazing systems that are impacted by PWDs (FEMA 2021).

Jobsites generally involve incomplete and unstructured resources that are susceptible to extreme wind events. In this regard, prior works such as Chavez (2016), Ham et al. (2017), Kamari and Ham (2021a, b), Kareem (1985), and Li and Ellingwood (2006) classified construction sites among the most vulnerable environments to extreme wind events such as hurricanes. Jobsites and laydown yards are used to temporarily store unsecured materials such as gravel pits, crushed rocks, and metal piping rods, that may become airborne during severe wind events, which may lead to substantial damage to neighboring environments. According to a report by the Department of Homeland Security (SunSentinel 2017), the aftermath of Hurricane Irma left more than 200,000 cubic yards of debris, which were primarily classified into construction-related materials and vegetation. The overall cost of debris removal was estimated as 1.4 billion dollars because the debris hauling process typically takes over several months to complete. To mitigate the impact of extreme wind events, state governments located in hurricane-prone areas have enacted ordinances that mandate superintendents, contractors, and safety directors to take proactive measures regarding hurricane preparedness in their jobsites (Bohm 2020). Disobedient contractors failing to implement proper measures to identify PWDs and secure them, are subject to fines and imprisonment of up to 60 days (Bohm 2020). For implementing such policies, hazard recognition in large-scale jobsites still depend on field observation by workers to recognize relevant hazards amidst irrelevant distractors (Jeelani et al. 2017). In this regard, despite substantial safety training programs, it has been known that

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Note. This manuscript was submitted on March 2, 2022; approved on May 25, 2022; published online on August 27, 2022. Discussion period open until January 27, 2023; separate discussions must be submitted for individual papers. This paper is part of the ***Journal of Computing in Civil Engineering***, © ASCE, ISSN 0887-3801.

inadequate hazard recognition and mitigation measures of workers contribute to up to 42% of the accidents involved in the industry (Haslam et al. 2005). Moreover, hazard recognition and preparedness measures are regrettably downplayed by workers as they are treated as secondary or not closely relevant to the primary task (Jeelani et al. 2017). To facilitate disaster preparedness, research efforts have leveraged a vision-based approach to detect, quantify, and analyze the risk of PWDs in construction sites (Ham and Kamari 2019; Kamari and Ham 2022). In this regard, unmanned aerial vehicles (UAVs) have enabled access to aerial vantage points to capture a record of hazardous objects such as PWDs (Kamari and Ham 2018; Liu et al. 2014). Despite their benefits, the focus of the prior works was primarily placed on the risk assessment of debris at jobsites, and there is a lack of studying the impact of such potential windborne debris on neighboring communities including buildings and residential environments. Given that glazing systems are among the most vulnerable elements of dwellings, knowledge about the jobsite debris-related risk is limited, and there is need for the reliability analysis of buildings and residential environments threatened by PWDs sourced from nearby jobsites.

In this paper, a novel computing method is proposed to assess the risk of PWDs in jobsites with respect to adjacent buildings in neighboring communities. In the proposed method, visual data analytics are performed based on virtual representation of communities and jobsite debris, and the following computing modules are proposed: (1) satellite imagery-based terrain modeling to study 3D characteristics of the at-risk built environment; (2) analyzing visual data from Google Street View to assess the risk associated with glazing panels of dwellings in the communities; and (3) analyzing visual data from a jobsite to quantify the amount of PWDs such as gravel, to associate its risk to neighboring communities. In this regard, by using images collected from jobsites, a 3D point cloud model is reconstructed, and PWDs (gravels in this paper) are automatically detected from the point cloud. Building on the semantic segmentation, vulnerable openings such as windows are analyzed in Google Street View images collected from built environment nearby jobsites, and the failure probability of glazing systems are computed building on the physical-based risk model by (FEMA 2021). Cases studies were performed to demonstrate the risk assessment of communities located nearby a railroad jobsite. The proposed framework has the potential to provide a heads-up for residents and communities that will be exposed to jobsite debris during extreme wind events. The residents impacted by PWDs from jobsites could take preemptive measures to mitigate the loss, which support establishing community resilience during extreme wind events.

Background

Visual Data-Driven Disaster Management

In past years, the world has experienced a record number of natural disasters, including hurricanes that caused billions of dollars in damage. In this regard, studies have been carried out toward proactive disaster management measures to reduce the cost and mitigate the loss imposed by such disasters. Generally, it has been known that every two or three dollars investment in disaster management could lead to up to one hundred dollars saved due to the disaster loss reductions (Pheng et al. 2006). The focus of prior works on disaster management are divided into pre- and postdisaster responses. The focus of predisaster management is primarily placed on adopting proactive measures to mitigate the impact of disasters, and the essence of postdisaster activities is set to implement the damage assessment to better plan recovery measures (Alizadeh et al. 2021). To rapidly assess the severity of damage after disasters such as hurricanes, remote sensing solutions, including ground/aerial images and 3D laser scanners, would be one of time- and cost-effective practices. In this regard, research efforts implement remote sensing techniques to perform postdisaster damage assessment (Guo 2010; Kashani et al. 2015; Olsen et al. 2013; Pesaresi et al. 2007; Tralli et al. 2005; Wang et al. 2010; Zhang et al. 2002). For instance, Li et al. (2008) and Ural et al. (2011) applied the LiDAR to measure landmark variances among pre- and posthurricane and tornadoes. In addition, aerial imagery captured in high (Vijayaraj et al. 2008), medium (Womble et al. 2006), and low (Wang and Xu 2010) spatial resolutions has been utilized for the damage assessment. Contextual information obtained from imagery was used to measure the normalized difference in vegetation index (NDVI) and assess damaged areas (Ill et al. 1997; Liou et al. 2010). To implement mitigation measures to reduce disaster risk in jobsites, it is required that practitioners such as safety managers accurately identify the hidden risk in a timely manner (Carter and Smith 2006). In this regard, eye-tracking technology has been proposed to study the hazard cognition in jobsites (Asadi et al. 2017; Jeelani et al. 2019). The focus of these works is placed on assessing the quality of hazard perception by practitioners and better training them to more effectively flag and analyze the extent of potential hazards and the associated threats. Due to the complexity of hazard analysis related to PWDs at different severity of wind events, they would not translate into sensible outcomes because factors such as physical and geometrical shape, unit-weight and wind speed can influence the debris behaviors and contribute to the complexity of the associated risk assessment (Wills et al. 2002). Recently, Kamari and Ham (2022) proposed the kinetic energy-driven risk assessment and visual data analytics using a deep learning technique to quantify the debris risk with respect to extreme wind events. Despite the potential in the context of disaster preparedness, the direction and the scope of the prior study was limited to given jobsites, and thus the vulnerability of nearby built environments including buildings and residential environments impacted by jobsite debris could not be encoded into the model.

Physical-Based Risk Assessment Models

Once objects (e.g., gravel) become airborne, they are transported in the three-dimensional wind field. The airborne grains will eventually collide with the ground or building façades and potentially cause a breach in the glazing systems. Modeling the behavior of such PWDs and predicting their potential impact have been studied for past years. In this regard, Kind and Wardlaw (1984) performed a wind tunnel experiment to investigate gravel scour patterns from rooftops that are due to spiral vortex flow. Later, Wills et al. (2002) classified the debris behavior, based on their physical shapes, into the three categories of compact-, rod-, and plate-type debris, and possible trajectories of the rod- and plate-type debris during severe wind events were studied by Holmes (2004) and Holmes et al. (2006). As such, there have been efforts to document the extent of damage to the glazing system, which was later used as a basis for examining windborne debris trajectory and damage models. For instance, during Hurricane Andrew, a high-rise complex building located in Kendall, in the state of Florida, suffered severe damage imposed by airborne gravel, and Behr and Minor (1994), Minor (1994), and T. Smith, personal communication, (1999) recorded photographs of glazing system damage. Later, these photos were used as a benchmark to assess the fidelity of the HAZUS debris model to estimate loss and damage to the glazing system (FEMA 2021). The HAZUS model on airborne gravel risk to

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buildings consists of two components (FEMA 2021). The first component provides a mathematical simulation of debris generation and transportation (e.g., trajectory). The gravel transport in the wind field is modeled by solving the equation of motion on gravel grains, and here the wind force acting on the gravel grains is the function of the time and location of the gravel (FEMA 2021). The second component builds on a risk model to approximate the probability of damage to building façades and glazing systems during extreme wind events. The risk model estimates the expected number of impacts and impact momentum by airborne gravel and approximates the failure chance of breach in façade components. In this research, we encode the proposed machine vision-based computing framework into the HAZUS risk model on airborne gravel, to study the risk of gravel in jobsites to nearby built environments.

Methodology

The computer vision-based reliability analysis in this paper builds upon the HAZUS model (FEMA 2021) by taking account of: (1) the amount of the source (i.e., windborne debris such as gravels), (2) the distance between the source and target buildings, (3) the windows of the target buildings, and (4) the wind speed. To carry out the risk analysis on the glazing system of buildings with respect to debris in nearby jobsites, the proposed framework consists of three computing modules, each with a particular type of visual data input (Fig. 1). First, building on satellite imagery, the terrain roughness is computed for the at-risk built environment, and the wind intensity in the given area is adjusted based on the terrain roughness. Second, images from façades of target buildings are collected from Google Street View, and the portion of the buildings occupied by the glazing is computed through the deep learning-based approach. Third, visual data from a jobsite is processed to quantify the volume of potential windborne debris (PWDs) in the jobsite (gravels in this paper). Given the debris volume, the number of expected debris impacts that are initiated from the jobsite is computed for each building, and building upon the fragility properties of the glazing systems, the associated reliability information (i.e., the probability of *no damage*) is computed as follows (FEMA 2021):

$$R = e^{[-\lambda q(1-P(\xi<\xi d))]} \tag{1}$$

where R = reliability of the glazing system threatened by the collision impact of jobsite debris. The parameter q is the portion of building façades occupied by the glazing computed by analyzing the outcome of semantic segmentation to detect building windows from Google Street View images. The parameter λ = number of expected debris impacts to the building, which is computed based on the HAZUS risk model (FEMA 2021). Based on the wind speed and the distance between target buildings and the gravel source, we build on the HAZUS model to analyze the number of expected debris impacts to the buildings during extreme wind events. In this research, the value of λ is adjusted based on the amount of gravel in a jobsite that is quantified through the proposed volumetric measurement framework. Finally, the term $P(\xi < \xi d)$ denotes the probability of glazing failure at a given momentum impact and wind speed. Because rough terrains reduce wind speed and the impact of debris, the characteristics of the terrain are analyzed based on the satellite image processing, and the wind reduction factor is computed accordingly. In the following section, each computing module is demonstrated and discussed in further detail.

Surface Roughness Modeling Based on Satellite Image Processing

Terrains with a rough characteristic could reduce the wind intensity at the immediate vicinity of the ground surface, but the intensity at the upper level remains almost the same. This is because vertical barriers (e.g., buildings, trees) are expected to dissipate the wind intensity in rough terrains. For example, the wind intensity experienced by two-story dwellings located in the middle of the forest might be as low as half of the wind intensity encountered by the same dwellings located in an open environment (FEMA 2021). As such, the density of trees and buildings needs to be taken into account for assessing the terrain roughness. Building upon Lettau's equation (Lettau 1969), the terrain roughness can be quantified as follows:

$$z_0 = 0.5\frac{HS}{A} = 0.5\frac{H_{bldg} \times S_{bldg} + H_{tree} \times S_{tree}}{A} \tag{2}$$

where z_0 = terrain roughness (m); H_{bldg} and H_{tree} = average height of buildings and trees in the terrain (m); and S_{bldg} and S_{tree} = demonstrate the total projected frontal area of buildings and trees (m^2), respectively. Parameter A = surface area of the target terrain (m^2). Due to the complexity of calculating the frontal area of trees, based on the HAZUS recommendation (FEMA 2021), the parameter S_{tree} could be estimated as follows:

$$S_{tree} = \alpha \times S_{bldg} \times 0.5 \tag{3}$$

where α = area occupied by trees over the area occupied by buildings in satellite imagery. In this paper, the terrain roughness is computed based on satellite image analysis. First, satellite images of built environments are collected from Google Maps. To assess the terrain roughness and locations of vertical barriers, the semantic segmentation is performed over the collected satellite images. Given satellite imagery [Fig. 2(a)], the pixel-level classification is performed to obtain the location of trees and buildings. The number of pixels classified as a tree is then divided into the number of pixels identified as a building, to compute the α parameter in the given built environment [Fig. 2(b)]. To calculate S_{bldg}, the semantic values corresponding to buildings are converted into the binary class [Fig. 2(c)]. A particular ID is assigned to each building boundary in the binary image. The projected frontal area of the buildings facing the wind direction is then calculated by measuring the distance between a maximum and minimum pixel of the building boundary in the y direction [e.g., p_{y1} and p_{y2} in [Fig. 2(d)], multiplied by the building height. Here, as a proof of concept, the height of buildings and trees are assumed to be uniform within the case study area. Building on the height information of buildings and trees, and frontal areas of the barriers, and the α parameter and areas of the given built environment (A), the terrain roughness is computed through the Latteus equation. To calculate the terrain roughness through satellite image processing with respect to various wind directions other than the x-axis direction, the binary image needs to be rotated accordingly before computing the projected frontal areas of the barriers.

At different building heights, the effective gust wind speed ratio is then calculated by taking account of the terrain roughness as demonstrated in Fig. 3, which is built upon the HAZUS model (FEMA 2021). As observed, for rougher terrains, the reduction factor of the wind speed is significant. By using the semantic segmentation (Fig. 2) and the relationship between gust wind speed ratio and surface roughness length by Lettau's equation, the terrain roughness with given building heights is computed.

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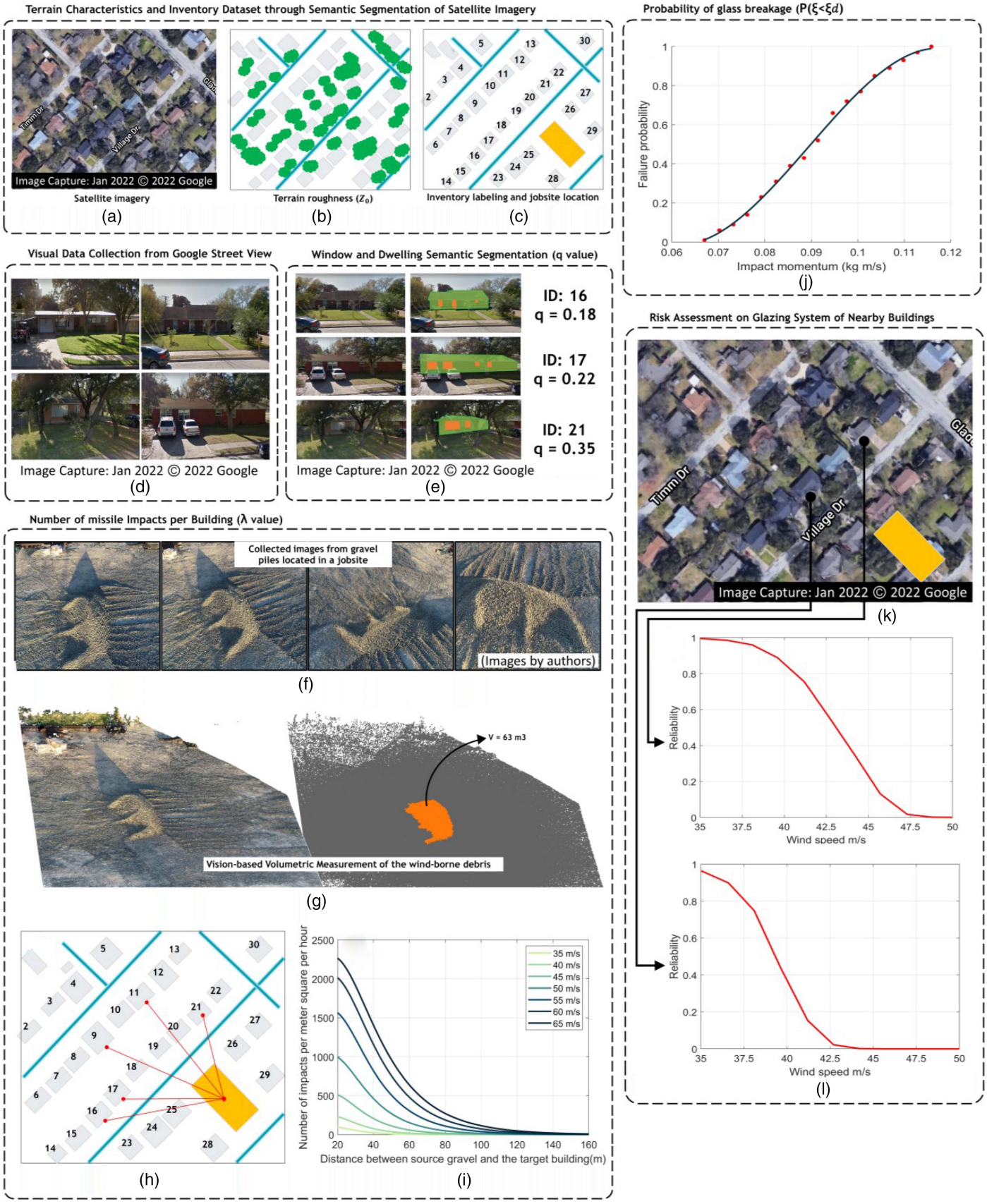


Fig. 1. Overview of the proposed method for analyzing safety risk imposed by jobsite debris to nearby built environments: (a) satellite imagery; (b) terrain roughness; (c) inventory labeling; (d) collecting selective Google Street images; (e) analyzing building façades with respect to glazing panels; (f) site imagery (images by authors); (g) volumetric measurement; (h) distance from the site; (i) number of gravel impacts; (j) analyzing the failure probability; (k) satellite imagery with a site location; and (l) reliability versus wind speed. [Images (a, d, e, and k) © 2022 Google.]

J. Comput. Civ. Eng., 2022, 36(6): 04022033

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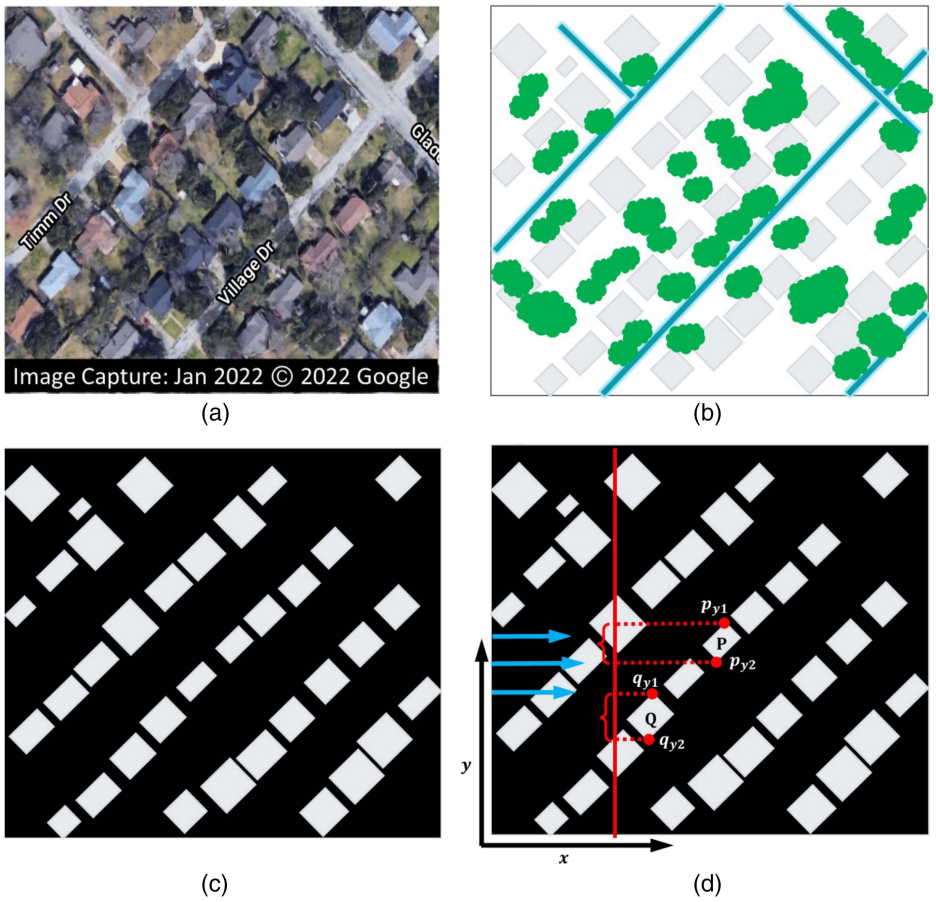


Fig. 2. Terrain roughness modeling through satellite image processing: (a) satellite imagery (image © 2022 Google); (b) semantic segmentation; (c) isolation of target objects; and (d) frontal area versus wind direction.

Analyzing Building Façades with Respect to Glazing Panels

The glazing system is among the most fragile component of dwellings becauses they can break due to the collision impact of PWDs. In this sense, buildings with larger openings and façades that are built from tempered glasses are more susceptible to break down during extreme wind events. In this regard, one of the key parameters in assessing the risk of glazing damage during extreme wind events is to quantify the portion of given buildings that are occupied by glass openings. In this research, we leverage publicly available large-scale visual data collected from Google Street View. To retrieve the image of building façades in given regions, an API request is established through an *HTTP* request in Google Street View (Balali et al. 2015; Chen et al. 2019; Kim et al. 2021). Multiple images from different perspectives are retrieved for target buildings [Fig. 4(a)]. The semantic segmentation is then performed, and the images with the best line of sight of the buildings are chosen for further processing. The outcome of semantic segmentation is studied among collected images, and the image containing the greater number of pixels classified under building and windows categories is selected to represent the best line of sight [Fig. 4(a)]. The semantic segmentation is carried out by a residual deep neural network, Resnet 50 (He et al. 2016; Thorpe and van Gennip 2018). The Resnet architectures demonstrate high performance in terms of accuracy and computation cost. Unlike other deep neural network architectures, the residual networks are robust for optimizing weights and generally gain accuracy upon increasing network layers. The size of residual networks is generally determined commensurate with the accessible computation power (He et al. 2016; Thorpe and van Gennip 2018). To train the Resnet network, we build upon the following four classes: building, windows, possible occlusion, and others. The class of possible occlusion consists of trees and cars that might occlude the line of sight to building façades. The other category involves pixels associated with sky,

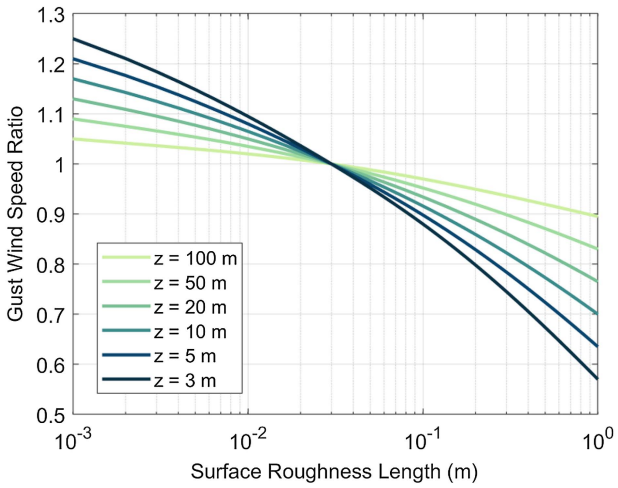


Fig. 3. Terrain roughness and gust speed ratio for different building heights.

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(a) (b)

Fig. 4. (a) Google Street View image retrievals; and (b) semantic segmentation for quantifying the parameter q of a building. (Images © 2022 Google.)

asphalt, or pavement. By using the outcome of semantic segmentation [Fig. 4(b)], the number of pixels classified as buildings and windows are analyzed, and the parameter q is calculated by dividing the number of pixels representing buildings over the number of pixels classified as windows. The parameter q in the HAZUS model incorporates the degree of vulnerability associated with the size of glazing panels into the HAZUS model. The parameter q ranges between zero and one, whereas the higher values indicate a higher vulnerability of the glazing system.

Expected Jobsite Debris Impacts to Buildings

Estimating the expected number of debris impacts to building exteriors such as windows is essential in identifying the reliability of glazing damage during extreme wind events. To obtain the expected number of debris impacts per building, there are three variables that are required to be considered: (1) distance between target buildings and the gravel source by analyzing satellite imagery (Module #1), (2) gust wind speed adjusted based on the terrain roughness (Module #1), and (3) volume of the gravel source in the jobsite by processing site images (Module #3). To quantify the number of missile impacts by PWDs to glazing systems, we build upon the following missile transport mechanism (FEMA 2021):

$$\lambda = N \times B \times W \tag{4}$$

where λ = number of expected debris impacts for the target building; N = number of expected debris impacts on the target building per unit area in 1 h; B = surface area of target building (m^2); and W = duration of the extreme wind event affecting the target building (h). To quantify the risk of the missile impact associated with gravels, the target PWDs in this paper, we build on a physical-based risk model that was validated based on wind tunnel experiments (Doddipatla and Kopp 2019). This approach established a relationship between the distance of the target building and gravel source versus the number of impacts in the unit area of the building within a time interval as follows:

$$\mathrm{N} = \mathrm{n}(\mathrm{V},\mathrm{d}) \times \mathrm{f}(\mathrm{d},\mathrm{h}) \tag{5}$$

$$\mathrm{n}(\mathrm{V},\mathrm{d}) = 53\frac{\left(1+\tanh\frac{\mathrm{V}^{1.11}-80}{18}\right)}{(e^{x}+e^{-x})}, \qquad \mathrm{x} = \frac{\mathrm{d}-15}{0.33\mathrm{V}} \tag{6}$$

$$\mathrm{f}(\mathrm{d},\mathrm{h}) = \frac{60}{\mathrm{h}_0}\cos^2\left(\frac{\pi}{2}\frac{\mathrm{h}}{\mathrm{h}_0}\right), \quad \text{if } \mathrm{h} \le \mathrm{h}_0,\ \mathrm{h}_0 = 0, \text{ or } \mathrm{h} > \mathrm{h}_0,\ \mathrm{h}_0 = 30 - \frac{\mathrm{d}}{10} \tag{7}$$

where N = number of expected debris impacts on a target building per unit area and per hour; V = peak gust speed (m/s) that can be computed based on surface roughness modeling described in the section of "Surface Roughness Modeling Based on Satellite Image Processing"; d = distance between the PWD and the target building (m) that can be computed from satellite image processing; and h = height of the target building (m). The number of impacts per unit area of the building within a time interval can be defined by two factors: the peak gust speed varying upon the distance of the target building [$\mathrm{n}(\mathrm{V},\mathrm{d})$] and the height of the target building [$\mathrm{f}(\mathrm{d},\mathrm{h})$]. Through Eq. (6), the speed factor can be increased as the distance between the PWD and the target building (d) is reduced, and the peak gust speed (V) is increased. The height factor also can be increased as the height of the building (h) and the distance between the PWD and the target building (d) are reduced based on Eq. (7). Fig. 5 demonstrates the number of impacts per unit area and per

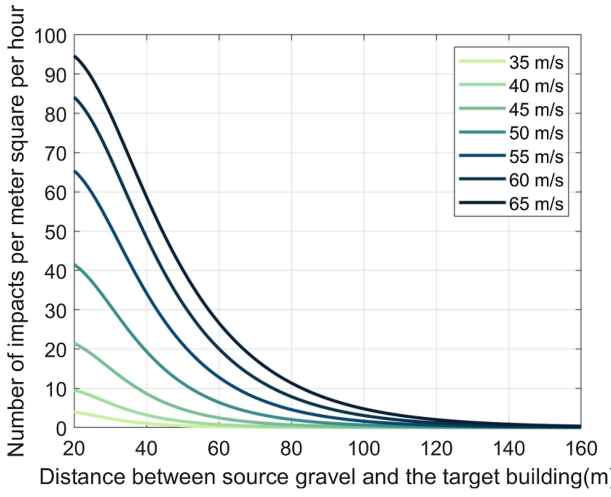


Fig. 5. Number of missile impacts per unit area and hour with respect to the distance from gravels (PWDs) at different wind speeds at 5 m height.

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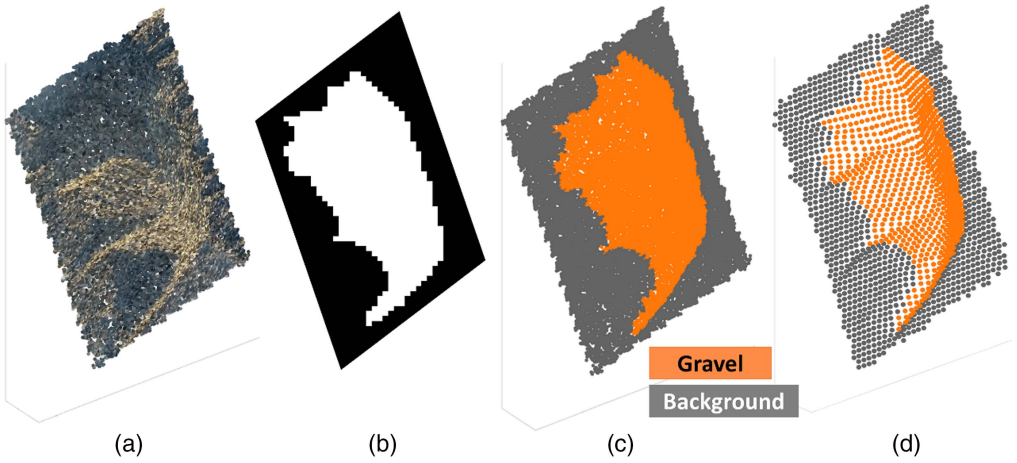


Fig. 6. Vision-based volumetric measurement: (a) a dense point cloud model; (b) the digital surface map of the point cloud model; (c) the binarization to detect the gravel pile at the 2D level; and (d) the projection of the 2D binarized segmentation onto the point cloud model.

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hour with respect to the distance between gravels and the target building at different wind speeds. As observed, the number of missile impacts declines when PWDs are located far from the target building. In addition, the number of impacts is increased at the higher level of gust speed due to the higher rate of gravel scour.

The relationship presented in Fig. 5 builds upon the HAZUS model based on a small-scale experimental setup. Compared with relatively large gravel volumes located in jobsites, the gravel volume used in the HAZUS model was about 3 m^3, and thus the number of impacts is required to be adjusted to be commensurate with the volume of gravels observed on the jobsite. Prior studies suggested that there is a correlation between the number of impacts and the airborne gravel volume (Doddipatla and Kopp 2019). Building on the experimental outcomes presented in the literature, in this study, the number of impacts is adjusted as follows:

$$N_{adjusted} = N \times \frac{V_{field}}{V_{empirical}} \tag{8}$$

where $N_{adjusted}$ = adjusted number of impacts per square meter and per hour; V_{field} = volume of gravel piles in a jobsite; and $V_{empirical}$ = volume of the gravel used in the HAZUS model. By obtaining $N_{adjusted}$, the parameter λ is computed using Eq. (4). In this research, building on Kamari and Ham (2022), V_{field} is computed by leveraging aerial and ground-level images of PWDs located in a jobsite. In this regard, (1) using collected images, a dense point cloud model is reconstructed [Fig. 6(a)], (2) the digital surface map of the point cloud model is generated [Fig. 6(b)] and binarized to detect the gravel pile at the 2D level, (3) the outcome of the 2D binarized segmentation is projected onto the point cloud model to detect the PWDs at the 3D level [Fig. 6(c)], (4) the segmented point cloud model is resampled [Fig. 6(d)], and the volumetric measurement is performed with the ground registration. The volume of a gravel pile builds upon the Z height of points in the resampled model that are identified as a gravel, multiplied by the square size of the grid (Kamari and Ham 2021c), which can be expressed as follows:

$$V_{grv} = GS^2 \times \sum_{i=1}^{n} \sum_{j=1}^{m} Z_{grv}((X_i, Y_j)) \tag{9}$$

where V_{grv} = volume of the gravel; Z_{grv} = height of points representing gravel at the location of (X_i, Y_j); and GS = discretization grid size. The number of grid elements at OX and OY directions are denoted as n and m. Fig. 6 illustrates the volumetric measurement framework.

Fragility Curves of Glazing Systems of Buildings

Exploring the characteristics of the glazing system is essential in damage prediction in extreme wind events. A dwelling with fragile glass is subjected to a higher failure risk than the one with tougher properties. In this research, building on the experimental outcomes of Dharani et al. (2004), the fragility of glass is analyzed based on laminated glass at different impact momentum. Laminated glass is a durable glazing system because it holds together once shattered. The profile of the specimen consists of outer glass ply, polyvinyl butyral interlayer, and inner glass ply. The glass specimen is subjected to 2 (g) steel balls at the speed of 20–50 m/s. Here, it is intended the small steel ball represents the gravel projectiles (Dharani et al. 2004). For each launch speed, the cumulative probability of the breakage is analyzed to obtain the fragility curve of the specimen (Fig. 7). The prior work (Dharani et al. 2004) experimentally found that when it comes to 2 g projectiles (gravel grain), wind speeds such as 25.9, 33.5, and 42.7 m/s generate 0.052, 0.067, and 0.085 kg · m/s impact momentum. Building on the experimental outcomes from Dharani et al. (2004), failure probabilities at these wind speeds are 0.11, 0.47, and 0.94, respectively. In this research, given a wind speed and the weight of the gravel grain, the momentum of impact is computed, and the failure probability of the glazing window system $P(\xi < \xi d)$ is obtained from this failure probability information at different impact momentum. The parameter ξ is the impact energy or momentum of the projectile, and ξd denotes the damage momentum threshold value.

Case Study

The at-risk built environment demonstrated in this paper is located nearby a railroad jobsite in the city of College Station, TX. The laydown yard associated with the jobsite involves a gravel pile. In general, the laydown yard of the railroad jobsite includes lumber, railway tracks, and substantial amounts of gravel, railroad ballast, and crushed stone piles (Raymond 2000; Strelkov et al. 2016). Gravels are used with or without any binders in railroad construction and function as the road base and foundation to hold railway tracks (Mengistu and Gebregziabher 2021; Tolppanen et al. 2002). The railroad project presented in this case study has the potential to

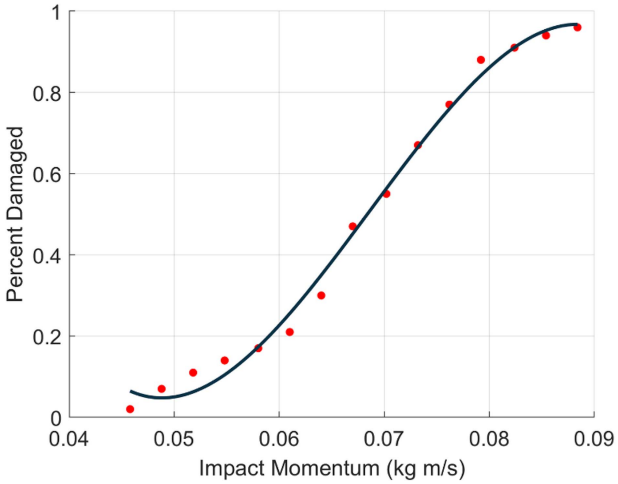


Fig. 7. Failure probability of a glazing system at different impact momentum.

threaten residential dwellings located at the immediate proximity of the jobsite. Due to the presence of trees and buildings in the given built environments, the terrain roughness and the associated gust wind reduction factor are computed through the proposed computing module. Satellite imagery of the terrain and the ground-level imagery of the dwellings are collected from Google Maps and Google Street View. To compute the volume of the gravel, aerial imagery was collected from the jobsite through a midend UAV. The average flight height of the UAV during the data collection was about 30 m, and the flight time for the aerial data collection from the jobsite was about 2 min. For the sake of simplicity in the point cloud segmentation, the aerial visual data were not collected from irrelevant objects (e.g., railway, tracks) in given areas. The proposed volumetric measurement is then performed to compute the volume of the gravel. To measure the accuracy of the volumetric measurement, experiments on small-scale gravel piles were separately performed, and the outcomes were discussed.

Terrain Roughness Modeling of the Case Study Area

To carry out the semantic segmentation, we built upon the large-scale annotated visual dataset provided by ISPARS (2020). By leveraging annotated images, a Resnet50 residual deep neural network is trained for the semantic segmentation of satellite images. The original satellite images (e.g., 6,000 px by 6,000 px) and their associated labels are divided into 100 patches with the size of 600 px by 600 px [Fig. 8(a)]. A total of 3,800 number of images were resized to 300 px by 300 px and were used to train the Resnet 50 network [Fig. 8(b)]. In this case study, the network is trained to detect buildings, trees, and other categories, in given satellite imagery. Fig. 8(c) demonstrates the outcome of the semantic segmentation based on the trained network.

Manual labels were provided on satellite imagery collected from the case study area. In this regard, a total number of 20 patches with the size of 300 px by 300 px were randomly selected and were semantically labeled with three classes: *Buildings*, *Trees*, and *Others*. The ground truth labels were used for evaluation; the mean accuracy for the segmentation is reported as 0.79, whereas the mean IoU and weighted IoU are identified as 0.49 and 0.79. The accuracy for detection of Buildings, Trees, and Others classes is reported as 0.73, 0.77, and 0.86, respectively.

By using the outcomes of the semantic segmentation of the satellite imagery and the height and projected frontal area of buildings and trees in given regions, the terrain roughness of the case study areas at different wind directions is analyzed [Fig. 9(d)]. The length of the arrows represents the roughness depending on the wind directions. Tompute the terrain roughness based on Eq. (2), it is assumed that the height of one-story buildings and trees are 6 m (WeekendBuilds 2022) and 10 m (Horticulture Unlimited Inc. 2022), respectively. The surface area of the target terrain is 30,322 m^2, which is shown in the red circle of Fig. 9.

Table 1 demonstrates the total projected frontal area of buildings and trees. Based on the semantic segmentation, the area occupied by trees over the area occupied by buildings [α in Eq. (3)] is obtained. Then, the projected frontal area of the trees is calculated accordingly. The surface roughness length (z_0) is computed using Eq. (2). Finally, the gust speed ratio is obtained building upon Fig. 3

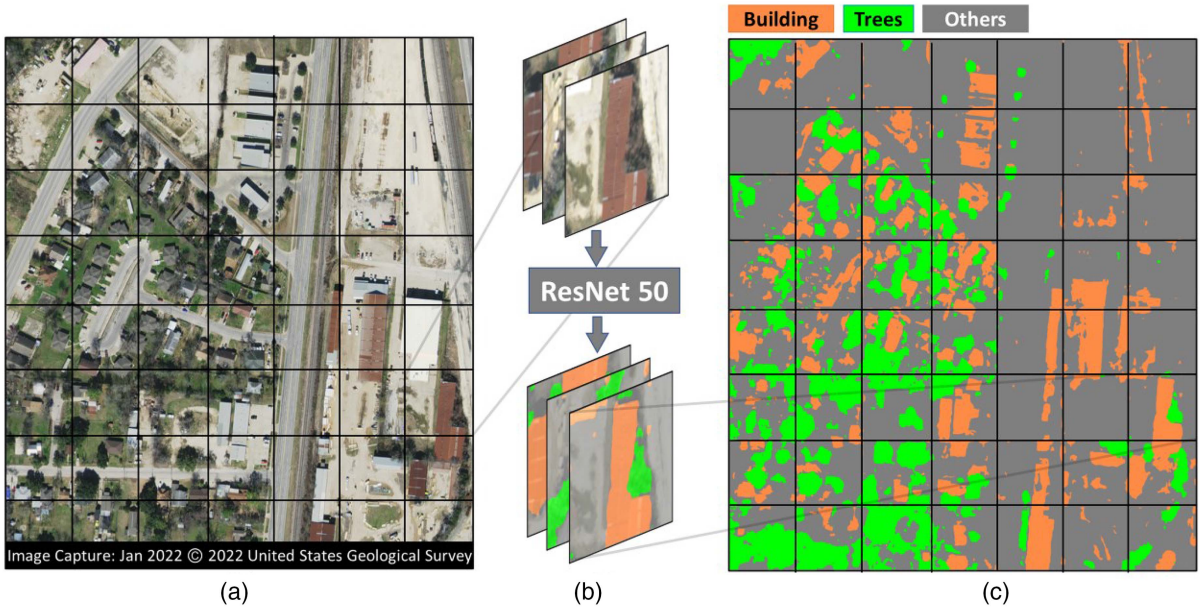


Fig. 8. Semantic segmentation for the case study area: (a) satellite imagery; (b) trained network for semantic segmentation; and (c) semantic segmentation. [Images (a and b) © 2022 Google, United States Geological Survey.]

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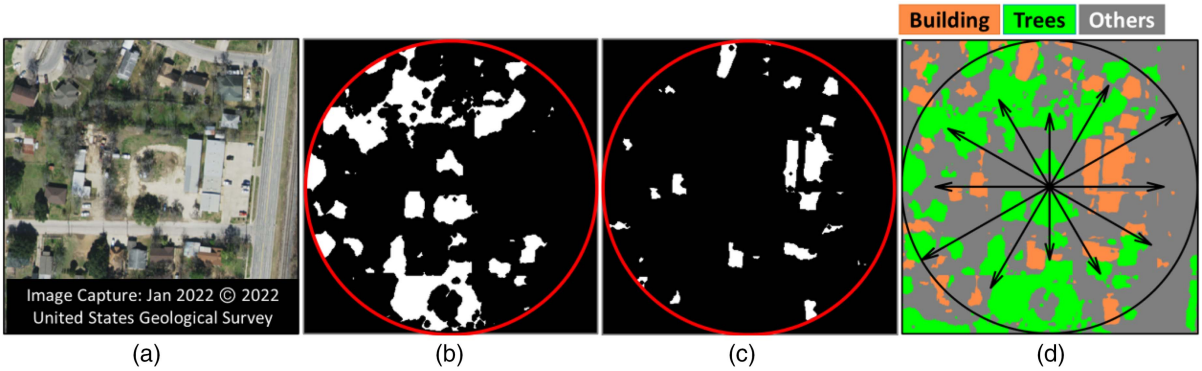


Fig. 9. Terrain roughness at different wind directions: (a) satellite image of the case study area (image © 2022 Google, United States Geological Survey); (b) segmented trees; (c) segmented buildings; and (d) surface roughness at different wind directions.

when z is 6 m; that is the average height of a one-story building (WeekendBuilds 2022). As observed, the wind direction in a rougher terrain is subjected to a higher reduction coefficient.

Glazing Detection through Deep Convolutional Neural Network

A deep neural network is trained to perform the glazing detection. In total, 600 images with the size of 1,150 px by 1,150 px, were collected from building facades using Google Street View, and they were annotated into windows, building, occlusion, and other categories (Fig. 10). Fig. 11 demonstrates the number of pixels associated with each category and the normalized frequency. 300 images were collected from the city of College

Table 1. Gust wind speed ratios at different wind directions ($\alpha = 1.56$)

Degree	S building (m^2)	S tree (m^2)	z_0	Gust wind speed ratio
0	2,216.7	1,731.5	0.50	0.73
30	2,237.2	1,744.3	0.50	0.73
60	2,146.1	1,670.0	0.48	0.74
90	1,981.5	1,547.8	0.45	0.75
120	2,240.2	1,746.6	0.50	0.73
150	2,416.6	1,880.5	0.54	0.72
180	2,216.7	1,731.5	0.50	0.73
210	2,237.2	1,744.3	0.50	0.73
240	2,146.1	1,670.0	0.48	0.74
270	1,981.5	1,547.8	0.45	0.75
300	2,240.2	1,746.6	0.50	0.73
330	2,416.6	1,880.5	0.54	0.72

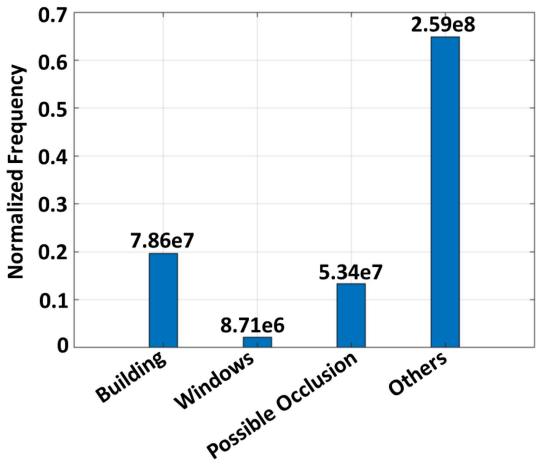


Fig. 11. Number of annotated pixels for each category (top of bars) and the normalized frequency in training datasets.



Fig. 10. Examples of annotated images with respect to semantic segments. (Images © 2022 Google.)

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Table 2. Performance of convolutional deep neural networks in the case study

Networks	Depth (layers)	Parameters (millions)	Averaged accuracy (%)
Alexnet	8	61.0	72.4
Vgg19	19	144.0	76.5
Resnet18	50	11.4	83.2
Resnet 50	101	25.6	85.0

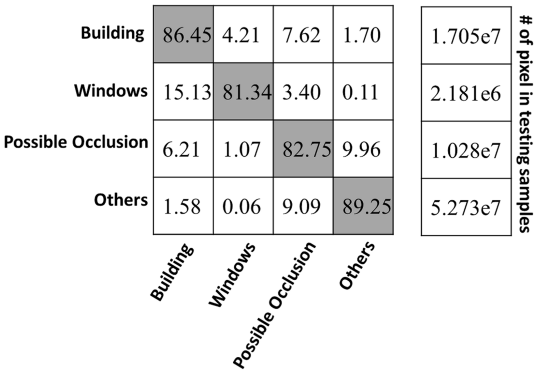


Fig. 12. Confusion matrix and sample size among semantic categories.

Station, TX, and the rest of the 300 images were captured from Houston, TX.

Different deep learning models were trained to carry out the semantic segmentation, including Alexnet (Krizhevsky et al. 2017), Vgg19 (Russakovsky et al. 2015), Resnet18 (He et al. 2016), and Resnet 50 (He et al. 2016). In this experiment, the fully convolutional network (FCN) semantic segmentation model (Long et al. 2015) was adapted for the Alexnet. For Vgg19 and Resnet 18 and 50, we trained a semantic segmentation model referred as Deeplab v3+ (Chen et al. 2018), which is identified under convolutional neural networks (CNN). For the networks, 450 images were used for training, and 150 images were used to evaluate the model. The characteristics of the networks, and the average accuracy of the semantic segmentation, are demonstrated in Table 2. The confusion matrix is shown in Fig. 12. It was observed that the confusion between possible occlusion and other categories is higher compared with the others due to textural similarities among these two categories. Based on the Resnet 50 model, the intersection over union (IoU) measure for Building, *Windows*, *Possible Occlusion*, and Others classes are 0.78, 0.58, 0.52, and 0.87, respectively. The mean IoU and weighted IoU for four classification categories are 0.69 and 0.8. Weighted IoU specifies the average IoU of each class that is weighted by the number of pixels associated with the class. By using the semantic segmentation outcomes, the parameter q is computed for the dwellings in the case study area, as shown in Fig. 13. Higher q represents a higher portion of the building occupied by vulnerable glazing systems.

Despite the benefit of using Google Street View images in the proposed method to assess the risk of jobsite debris in residential areas, it is still challenging to analyze the condition of building glazing systems in given regions because other objects might occlude the objects of interest. Such occlusion triggers a false negative detection of windows, as shown in Fig. 14. The semantic class shown in red is the possible occlusion category that contributes to the computing error in the q value. Such error associated with the occlusion is divided into two categories as follows: (1) as windows are occluded, and thus, the lower q values could be inaccurately obtained; and (2) when buildings are occluded, the q values could be overestimated. In this regard, the proposed method is expected to demonstrate poor performance for communities having large numbers of cars, trees, and shrubs. The information regarding occluded buildings would be retrieved based on similarities among buildings in a neighborhood such as Wang et al. (2021). For instance, similarities in façade features would be considered for inferring the q parameter among buildings. These remain as open research challenges and will be addressed in our future work.

Volumetric Measurement through Geometric Digital Twins

45 aerial images with the size of 4,096 px by 2,160 px were collected from the jobsite. First, a dense point cloud model is reconstructed, and then a binarized digital surface map is generated to

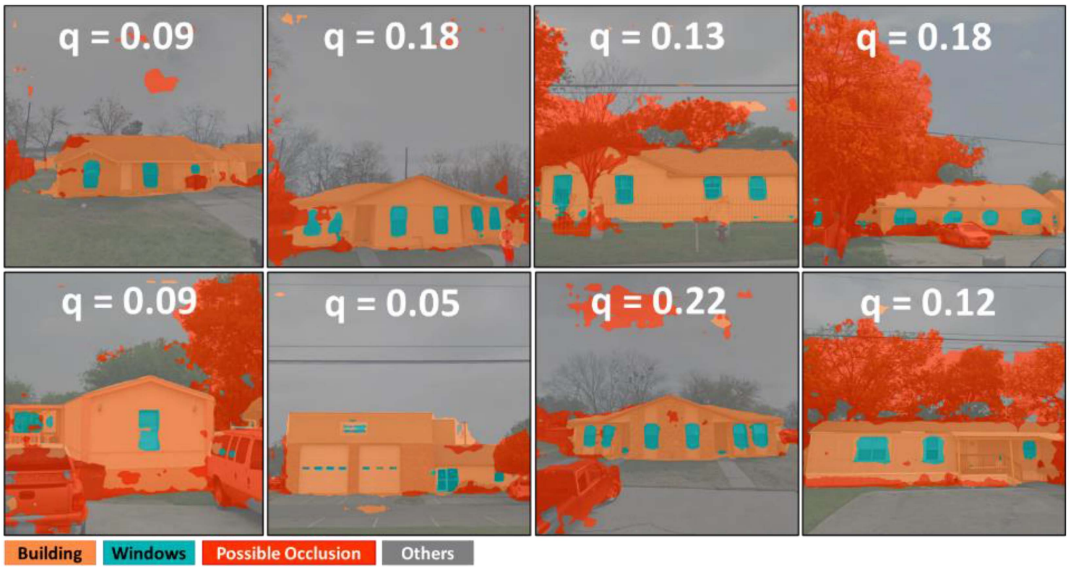


Fig. 13. Computing the parameter q based on the semantic segmentation.

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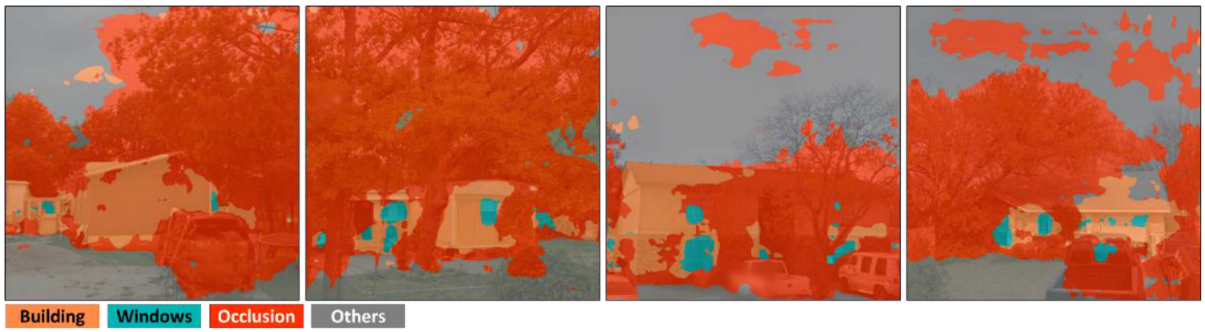


Fig. 14. Failure cases of detecting windows due to occlusion.

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detect the gravel object at the 2D level. The outcome of 2D detection is projected back onto the point cloud model to obtain the 3D gravel pile segmentation (Fig. 15). The point cloud model is resampled, and the ground is registered to compute the volume. In this experiment, the volume of jobsite gravels was computed as 63 m^3.

To identify the number of expected jobsite debris impacts for target buildings (λ), we estimate the adjusted number of impacts per square meter per hour ($N_{adjusted}$) based on the behavior of windborne debris from the wind tunnel test (Wills et al. 2002). Building on Eq. (5), the N value was estimated, varying with the distance between the PWD and the target building (d). To consider the effect on the scale of the experiment, the adjusted number of impacts per square meter per hour ($N_{adjusted}$) needs to be computed based on the volume of gravels that would affect the target building. For example, when d is 80 m and the wind speed is 62 m/s, the number of impacts (N) is approximately 10 in the small-scale experiment, as shown in Fig. 5. In small-scale experiments, the volume of gravel flying away ($V_{empirical}$) was 2.65 m^3, and the volume of a gravel pile in the jobsite of our case study (V_{field}) is 63 m^3. Accordingly, $N_{adjusted}$ becomes ~237 when wind speed is 62 m/s, and the distance between the target building and PWDs is 80 m. Fig. 16 demonstrates the adjusted number of impacts per unit area per hour, with respect to the distance between gravels and the target building at different wind speeds. Based on the estimated $N_{adjusted}$ along with the surface area of the target building and the duration of the extreme wind events affecting the target building, the impact of damage on given built environments is computed depending upon the distance between PWDs and target buildings [Eq. (4)].

Meanwhile, a benchmark dataset was collected to evaluate the volumetric measurement of gravel piles. A total of 39 images was collected from a gravel pile with a ground-truth volume of 12,568 cm^3. There was at least 30 percent overlap among collected

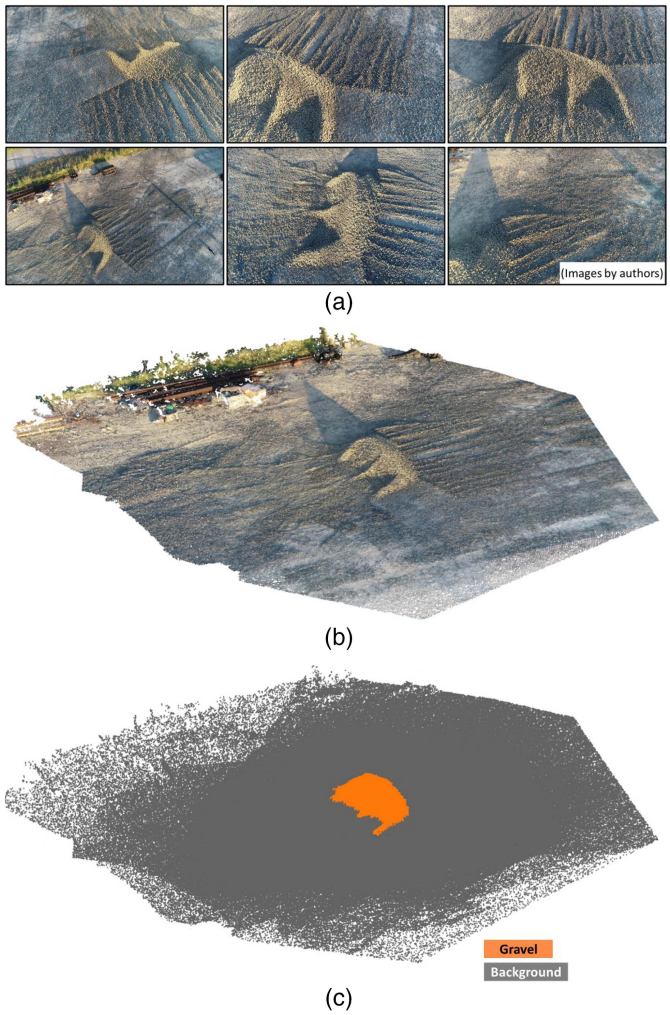


Fig. 15. (a) Examples of collected aerial images from a jobsite; (b) reconstructed point cloud of a gravel pile; and (c) segmented point cloud model. (Images by authors.)

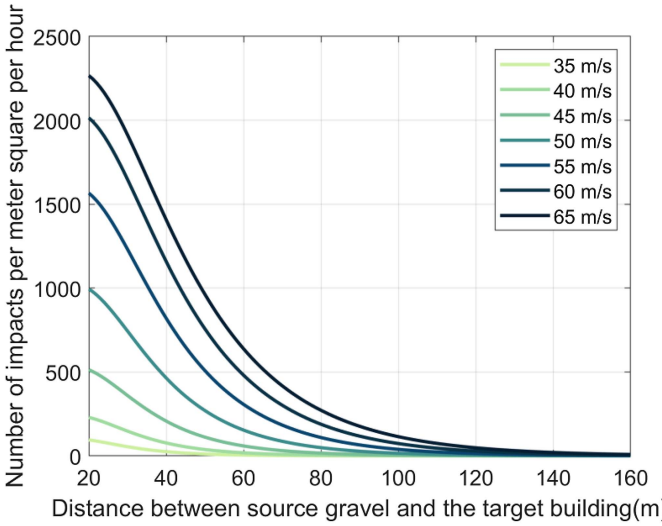


Fig. 16. The adjusted number of missile impacts per unit area and hour for the case study.

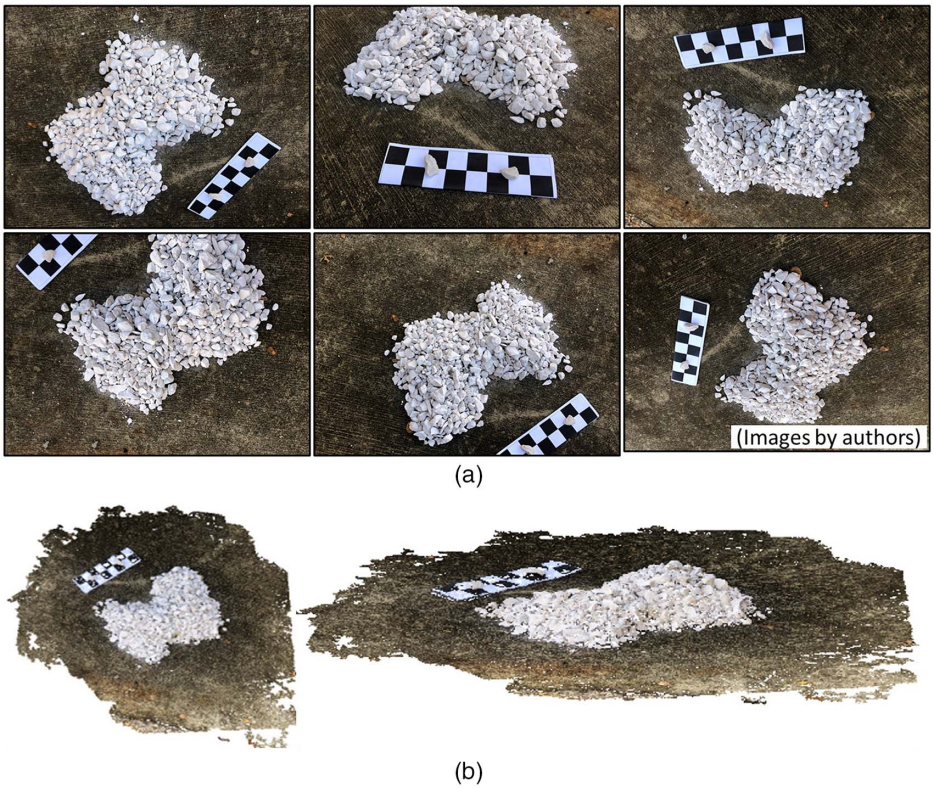


Fig. 17. (a) Collected images; and (b) the reconstructed point cloud model for benchmark dataset. (Images by authors.)

images to ensure a successful point cloud reconstruction. Fig. 17 demonstrates the collected images and reconstructed point cloud model. A marker was used as a reference to register the point cloud model to the real-world coordinate system. To perform the volumetric measurement, the point cloud model was resampled with grid sizes of 1, 1.5, and 2 cm, and the accuracy of volumetric measurement is reported in Table 3. As observed, the averaged error margin (EM) for the volumetric measurement was approximately 4.6%. The results infer that the grid size as a hyperparameter does not substantially impact the outcome of the volumetric measurement, because the measurements were close to the mean value with a standard deviation of 0.75.

Similarly, the volumetric measurement of the gravel pile in our real-world case study is illustrated in Fig. 18. Lower grid size values resulted in a finer resampled model and could slightly improve the accuracy of the volumetric measurement by sacrificing the computation cost. Meanwhile, the computational complexity with a larger grid size is lower, and the accuracy is expected to be slightly compromised. We observed that the standard deviation for the volumetric measurement among different grid sizes was 0.16, which indicates that the grid size does not significantly influence the performance of the vision-based volumetric measurement of gravel piles.

Table 3. Vision-based volumetric measurement versus the grid size

Grid size (cm)	Volumetric measurement (cm^3)
1	12,078, EM = −3.8%
1.5	11,978, EM = −4.7%
2	11,891, EM = −5.3%

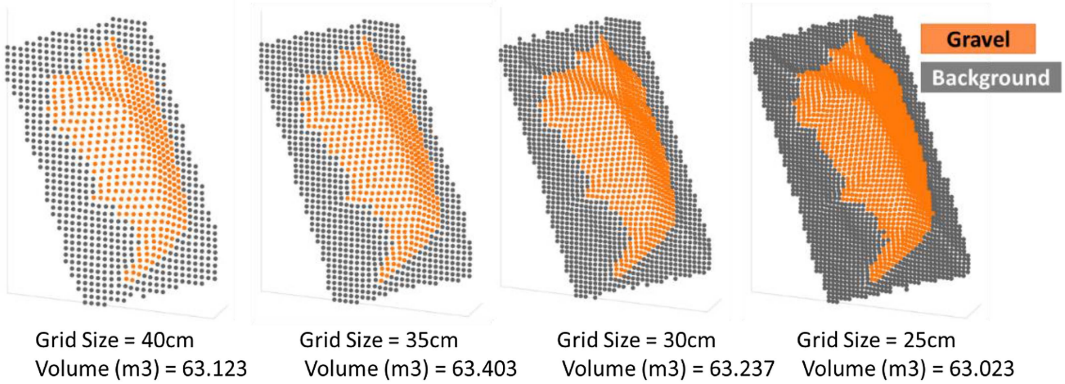


Fig. 18. Volumetric measurement of a gravel pile with different grid sizes in the case study.

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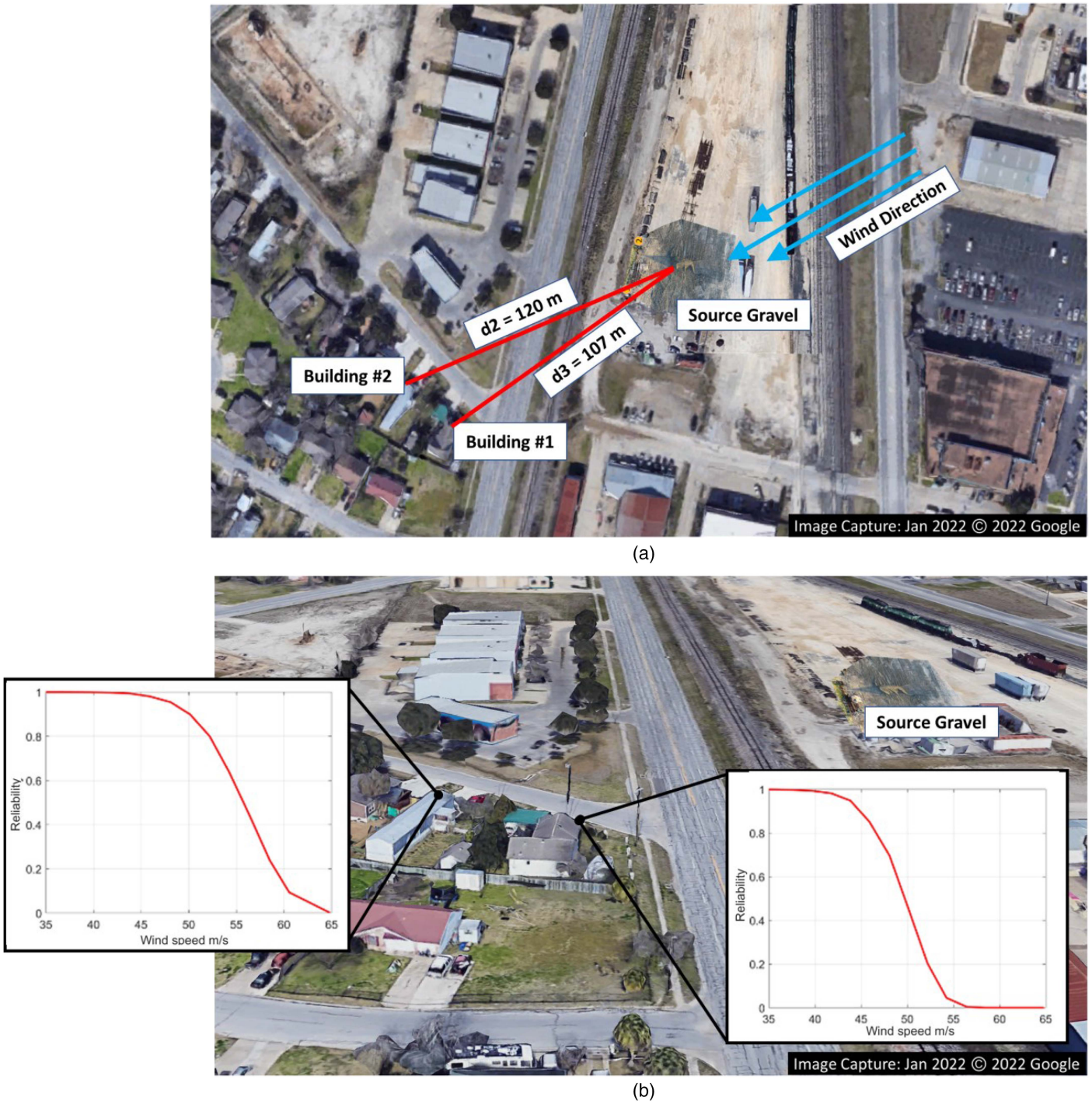


Fig. 19. (a) Case study area; and (b) reliability information (i.e., the probability of *no damage*) of building glazing systems located nearby the jobsite in the case study. (Images © 2022 Google.)

Reliability Analysis of Building Glazing Systems

By adjusting the wind speed based on the terrain roughness modeling, computing the parameters λ, q, and taking account of the property of the glazing system $P(\xi < \xi d)$, the reliability of the glazing system can be computed using Eq. (1). In the case study, it is assumed that the window is to be annealed with a thickness of 3.18 mm, and the gravel grains from the jobsite are consistent in diameter (12 mm) and weight (2 g). The point cloud reconstructed for the gravel is registered into the image, and the distance between target buildings and the source gravel is calculated. Examples of the distances between target buildings and the source gravel are shown in Fig. 19, and the q values of the building and the reliability curves of the dwellings located nearby the jobsite with respect to 35–65 m/s wind speed at a SW direction (225°) is demonstrated in Fig. 19. The gust wind speed ratio is calculated as 0.73 based on the terrain analysis.

Discussion

FEMA presented a simplified physical-based risk model, referred to as the HAZUS model, to predict the risk of the glazing system

damage during extreme wind events (FEMA 2021). The HAZUS model provides a reliable and robust framework to estimate the severity of loss in the glazing system with respect to the wind speed (FEMA 2021). There are assumptions in the HAZUS risk model to assess glazing system damage. For example, (1) the wind direction is persistent and is not changed during the wind event, (2) the property of the windows is considered to be uniform among housing, and (3) airborne gravels are uniform in size and mass. In this research, visual analytics was integrated with the HAZUS model to quantify the risk imposed by jobsite debris in glazing systems of the nearby housing. Despite its benefits in providing estimates of the potential wind-induced damage, the HAZUS model in the proposed work needs to be further discussed as follows.

Surface Roughness Estimation

An accurate estimate of the surface roughness of a landscape substantially contributes to effectively predicting wind-induced damage and loss. The Lettau's equation offers a means to quantitatively predict surface roughness from satellite images. However, in accordance with the FEMA recommendation, the use of Lettau's equation is subjected to a thorough engineering judgment in approximating the height of buildings and trees (FEMA 2021). In this regard, significant uncertainties might exist in estimating surface roughness in terrains where trees are the major obstacles. In this research, we assume the number of floors of buildings in the case study area to obtain the common building height. Accordingly, the height of the trees was estimated relative to the building heights that are presented in the case study area. In practice, similar investigations need to be conducted to satisfy the FEMA's recommendation to quantify the height and frontal areas of the buildings and trees.

Challenges in Vision-Based Building Façade Analysis

Although visual data collection from all sides of building façades has an indisputable advantage in the context of the completeness of the risk analysis, it is not trivial to leverage such data for the given regions and neighborhoods at scale. To broaden the impact and applicability of the proposed vision-based risk analysis at scale, visual analytics of buildings in the proposed method is limited to the part of the façade that is facing the street due to the nature of visual data collection building on Google Street View, similar to the prior work for assessing the seismic vulnerability of masonry walls using images collected from the street nearby buildings (Wang et al. 2021). It is assumed that the frontal façade facing the street involves a large portion occupied by the glazing system, which is likely vulnerable to extreme wind events that needs to be analyzed in this module.

Challenges in Evaluating the Risk Model

Although the proposed probabilistic approach can provide situational awareness of how jobsite debris could affect nearby built environments in extreme wind events, further research to validate the fragility analysis with real disaster cases will help compare the analytical results and real data. Although the proposed framework based on the FEMA's risk model is expected to reliably predict the imposed damage to the glazing system, there are still challenges in evaluating FEMA's risk model in real-world scenarios. For instance, factors such as the volume of jobsite windborne debris and distance between debris and target buildings are closely relevant to the potential damage to the glazing system. However, the locations of specific debris are not typically identified, and their volumes are not proactively quantified in real-world wind events. Lack of such information (e.g., volume, type, and location of debris) poses an immediate challenge to properly associate the observed damage with the volume/location of specific debris, and compare real damages with prediction outcomes. In addition, FEMA's probabilistic risk model may require large-scale observations for validation and potential mathematical refinement. In this regard, the challenge regarding the size of data to effectively evaluate the FEMA model based on real-world scenarios has been presented, featured, and highlighted by prior studies such as Johnson and Mahoney (2011). Further research to validate the findings with real disaster cases will discuss: (1) what are the required data to investigate the cause of failures of glazing systems of buildings nearby jobsites in extreme wind events?; and (2) how can we robustly collect such data at-scale?

Conclusion

The aftermath of extreme wind events such as hurricanes leaves behind billions of dollars in damage every year. Whereas structural damage and inundations are common occurrences, the majority of the damage is imposed by the collision impact of windborne debris. To this end, jobsites having gravel piles and aggregates are classified among the most hazardous environments to neighboring communities. During extreme wind events, windborne debris from jobsites can impose a devastating effect on nearby buildings and their associated glazing systems such as windows and glass openings. For the associated reliability analysis, a novel vision-based computing method is presented in this paper. The proposed method builds upon visual data at three fronts: (1) satellite images collected through Google Maps from the at-risk built environment threatened by jobsite debris, (2) ground-level images of building windows and glazing systems that are collected through Google Street View, and (3) visual data collected using UAVs from jobsites for geometric digital twinning to quantify the volume and impacts of jobsite debris. Being able to identify the hidden risk of construction materials imposed on nearby communities would allow practitioners to better secure their jobsites before extreme wind events and thus help mitigate the damage and loss during extreme wind events. Despite the benefits of the proposed method in the context of disaster preparedness, there are still open research challenges as follows: (1) ground-level images collected from building facades could be occluded by cars and trees, which may limit properly study the condition of building glazing systems; and (2) the digital twin framework to assess jobsite debris through aerial imagery might demonstrate poor performance in suboptimal weather conditions such as rain (Porav et al. 2019) or fog (Sakaridis et al. 2018). The aforementioned challenges are currently being explored as part of our ongoing research.

Data Availability Statement

Some or all data, models, or code (e.g., site images, training models) that support the findings of this study are available from the corresponding author upon reasonable request.

Acknowledgments

This material is based upon work supported by the National Science Foundation under Grant No. 1832187. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation.

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